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Prevost applied the idea of 'thermal equilibrium' to radiation. He suggested that all bodies radiate energy but hot bodies radiate more heat than the cooler bodies. At one point of time the rate of exchange of heat from both the bodies will become the same. Now the bodies are said to be in 'thermal equilibrium'.

Only at absolute zero temperature a body will stop emitting. Therefore *Prevost theory states that all bodies emit thermal radiation at all temperatures above absolute zero irrespective of the nature of the surroundings.*

8.3.2 Stefan Boltzmann law

Stefan Boltzmann law states that, *the total amount of heat radiated per second per unit area of a black body is directly proportional to the fourth power of its absolute temperature.*

$$E \propto T^4 \text{ or } E = \sigma T^4 \qquad (8.10)$$

Where, σ is known as Stefan's constant. Its value is 5.67×10^{-8} W $m^{-2} k^{-4}$

Note

If a body is not a perfect black body, then

$$E = e \sigma T^4 \qquad (8.11)$$

Where 'e' is emissivity of surface.

Emissivity is defined as the ratio of the energy radiated from a material's surface to that radiated from a perfectly black body at the same temperature and wavelength.

8.3.3 Wien's displacement law

In the universe every object emits radiation. The wavelengths of these radiations depend on the object's absolute temperature. These radiations have different wavelengths and all the emitted wavelengths will not have equal intensity.

Wien's law states that, *the wavelength of maximum intensity of emission of a black body radiation is inversely proportional to the absolute temperature of the black body.*

$$\lambda_m \propto \frac{1}{T} \text{ (or) } \lambda_m = \frac{b}{T} \qquad (8.12)$$

Where, b is known as Wien's constant. Its value is 2.898×10^{-3} m K

It implies that if temperature of the body increases, maximal intensity wavelength (λ_m) shifts towards lower wavelength (higher frequency) of electromagnetic spectrum. It is shown in Figure 8.13

Graphical representation

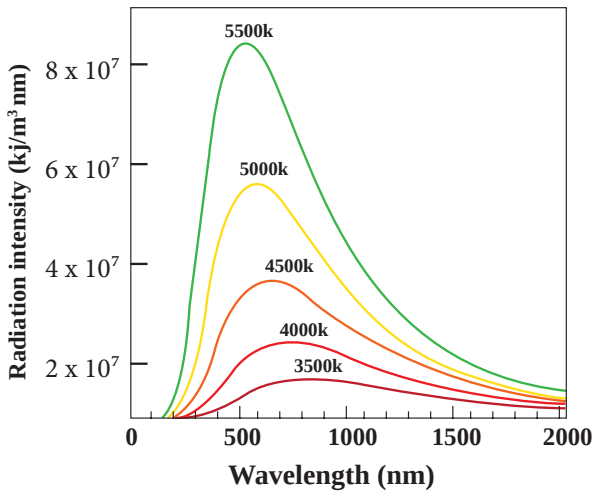


Figure 8.13 Wien's displacement law

From the graph it is clear that the peak of the wavelengths is inversely proportional to temperature. The curve is known as 'black body radiation curve'.

Wien's law and Vision:

Why our eye is sensitive to only visible wavelength (in the range 400 nm to 700nm)?